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$\exists!\mathbf{w} \in V : T(\mathbf{v}) = \langle \mathbf{v}, \mathbf{w} \rangle$

Let $\mathbf{u}_1, ..., \mathbf{u}_m$ be an orthonormal basis of V.

$$T : V \to \mathbb{R}$$

$$\begin{aligned} \mathbf{w} &= \sum_{i=1}^{m} T(\mathbf{u}_i)\mathbf{u}_i \\ &= T(\mathbf{u}_1)\mathbf{u}_1 + \cdots + T(\mathbf{u}_m)\mathbf{u}_m \end{aligned}$$

Let $\mathbf{v} = a_1\mathbf{u}_1 + \cdots + a_m\mathbf{u}_m \in V$, then $T(\mathbf{v}) = a_1T(\mathbf{u}_1) + \cdots + a_mT(\mathbf{u}_m)$. Finally

$$\begin{aligned} \langle \mathbf{v}, \mathbf{w} \rangle &= a_1\langle \mathbf{u}_1, T(\mathbf{u}_1)\mathbf{u}_1 \rangle + \cdots + a_m\langle \mathbf{u}_m, T(\mathbf{u}_m)\mathbf{u}_m \rangle \\ &= a_1T(\mathbf{u}_1) + \cdots + a_mT(\mathbf{u}_m) \\ &= T(\mathbf{v}) \end{aligned}$$

$\mathrm{row}(A) \cap \mathcal{N}(A) = \{\mathbf{0}\}$ where $A \in R^n$

$A \in \mathbb{R}^{m \times n}$ with rows $A = (\mathbf{a}_1 \cdots \mathbf{a}_m)^T$. Denote by $t_{\mathbf{a}}(\mathbf{x}) = \langle \mathbf{a}, \mathbf{x} \rangle = a_1x_1 + \cdots + a_nx_n$. Then the corresponding map is

$$T_A = \begin{pmatrix} t_{\mathbf{a}_1} \\ \vdots \\ t_{\mathbf{a}_m} \end{pmatrix}$$

Let $\mathbf{x} \in \mathrm{row}(A) \cap \mathcal{N}(A)$ then

$$A\mathbf{x} = \begin{pmatrix} \langle \mathbf{a}_1, \mathbf{x} \rangle \\ \vdots \\ \langle \mathbf{a}_m, \mathbf{x} \rangle \end{pmatrix} = \mathbf{0}$$

but since $\mathbf{x} \in \mathrm{row}(A)$

$$\mathbf{x} = r_1\mathbf{a}_1 + \cdots + r_m\mathbf{a}_m$$

and $\langle \mathbf{a}_i, \mathbf{x} \rangle = 0 \Rightarrow r_1\langle \mathbf{a}_1, \mathbf{x} \rangle + \cdots + r_m\langle \mathbf{a}_m, \mathbf{x} \rangle = \langle \mathbf{x}, \mathbf{x} \rangle = 0$. Therefore $\mathbf{x} = \mathbf{0}$.

Linear Maps are Determined by Their Basis

$T : V \to W$, then $\mathbf{x} = r_1\mathbf{v}_1 + \cdots + r_n\mathbf{v}_n$ which is unique.

$$T(\mathbf{x}) = r_1T(\mathbf{v}_1) + \cdots + r_nT(\mathbf{v}_n)$$

Likewise using a similar analysis we get the Rank-Nullity Theorem (7.4)

$$\dim \ker(T) + \dim \mathrm{im}(T) = \dim V$$

V and W over the same field are isomorphic $\iff$ they have the same dimension

Choose basis $\mathbf{v}_1, ..., \mathbf{v}_n$ of V, and $\mathbf{w}_1, ..., \mathbf{w}_n$ of W.

Let $T : V \to W$ be the unique linear mapping $T(\mathbf{v}_i) = \mathbf{w}_i$ (guaranteed by above proposition).

So $\mathbf{w}_1, ..., \mathbf{w}_n \in \mathrm{im}(T)$ and hence T is surjective.

Also $\Rightarrow \dim \mathrm{im}(T) = \dim V$, and therefore $\dim \ker(T) = 0$. So T is injective.

Set of All Isomorphisms $GL(V)$

$GL(V)$ is the set of all isomorphisms $T : V \to V$.

If $V = K^n$, then $GL(V) = GL(n, K)$.

$$\mathcal{N}(A^TA) = \mathcal{N}(A) \qquad \forall A \in \mathbb{R}^{n\times n} \Rightarrow \operatorname{rank}(A^TA) = \operatorname{rank}(A)$$

$$(A^TA)b = \begin{pmatrix} \langle a_1, a_1\rangle \dots \langle a_1, a_n\rangle \\ \vdots \\ \langle a_n, a_1\rangle \dots \langle a_n, a_n\rangle \end{pmatrix} b = 0$$

for all i

$$\langle a_i, r_1a_1 + \cdots + r_na_n\rangle = 0$$
$$\Rightarrow \mathbf{x} = (r_1 \cdots r_n)^T \in \mathcal{N}(A)$$

TS is an Isomorphism $\iff$ S injective, T surjective, and $\dim U = \dim W$

U, V, W are finite-dimension vector spaces over a field K.

$$S : U \to V$$

$$T : V \to W$$

are linear.

TS is injective $\iff$ S is injective and $\operatorname{im}(S) \cap \ker(T) = \{\mathbf{0}\}$

TS is injective so $\ker TS = \{\mathbf{0}\}$. Restricting T, we see

$$S : U \to V$$

$$T : \operatorname{im}(S) \to W$$

that both must be injective, so that $\ker S = \{\mathbf{0}\}$ and

$$\ker(T) \cap \operatorname{im}(S) = \{\mathbf{0}\}$$

TS is surjective $\iff$ T is surjective and $V = \operatorname{im}(S) + \ker(T)$

For $TS : U \to V \to W$ to be surjective, obviously T is surjective.

Now since TS is surjective, then $T(\operatorname{im}(S)) = W$, and $V = \operatorname{im}(S) + \ker(T)$.

TS is an isomorphism $\iff$ S is injective, T is surjective, and $\dim(U) = \dim(W)$

S is injective and finite dimension, means it is also surjective and an isomorphism. Likewise for T which is surjective. We can see this by letting $\phi : A \to B$ then

$$\dim A = \dim \ker(\phi) + \dim \operatorname{im}(\phi)$$

$$U \xrightarrow{S} V \xrightarrow{T} W$$

are isomorphisms. So $\ker(S) = \{\mathbf{0}_U\}, \ker(T) = \{\mathbf{0}_V\}$. So $\dim \ker(S) = \dim \ker(T) = 0$, and $\dim \operatorname{im}(S) = \dim(V), \dim \operatorname{im}(T) = \dim(W)$.

Using the Rank-Nullity theorem, we see $\dim(U) = \dim(V)$ for S and $\dim(V) = \dim(W)$ for T.

Isometries and Orthogonal Linear Maps

Let $S : V \to V$ be a map then $\forall \mathbf{a}, \mathbf{b} \in V$, we have

- *Isometry*: $d(\mathbf{a}, \mathbf{b}) = |\mathbf{a} - \mathbf{b}| = d(S(\mathbf{a}), S(\mathbf{b}))$ and S is a linear map.
- *Orthogonal*: $\langle S(\mathbf{a}), S(\mathbf{b})\rangle = \langle \mathbf{a}, \mathbf{b}\rangle$

Orthogonal Maps are Linear

We have to show $S(\mathbf{a} + \mathbf{b}) = S(\mathbf{a}) + S(\mathbf{b})$ which is equivalent to $|S(\mathbf{a} + \mathbf{b}) - S(\mathbf{a}) - S(\mathbf{b})|^2 = 0$. Simply expand this out, and use the fact $\langle S(\mathbf{x}), S(\mathbf{y})\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$.

Orthogonal Maps are Isometries

$$\begin{aligned}|S(\mathbf{b}) - S(\mathbf{a})|^2 &= |S(\mathbf{b}-\mathbf{a})|^2 \\ &= \langle S(\mathbf{b}-\mathbf{a}), S(\mathbf{a}-\mathbf{b})\rangle \\ &= \langle \mathbf{b}-\mathbf{a}, \mathbf{b}-\mathbf{a}\rangle \\ &= |\mathbf{b}-\mathbf{a}|^2\end{aligned}$$

Isometries are Orthogonal Maps

We have $|\mathbf{x}-\mathbf{y}|^2 = |S(\mathbf{x}) - S(\mathbf{y})|^2$ and so $|\mathbf{x}|^2 = |S(\mathbf{x})|^2$

$$\begin{aligned}|\mathbf{a}-\mathbf{b}|^2 &= (\langle \mathbf{a}, \mathbf{a}\rangle + \langle \mathbf{b}, \mathbf{b}\rangle) - 2\langle \mathbf{a}, \mathbf{b}\rangle \\ &= (\langle S(\mathbf{a}), S(\mathbf{a})\rangle + \langle S(\mathbf{b}), S(\mathbf{b})\rangle) - 2(\mathbf{a}, \mathbf{b}\rangle\end{aligned}$$

but $\langle \mathbf{a}-\mathbf{b}, \mathbf{a}-\mathbf{b}\rangle = \langle S(\mathbf{a}) - S(\mathbf{b}), S(\mathbf{a}) - S(\mathbf{b})\rangle$ so we get the result

$$2\langle \mathbf{a}, \mathbf{b}\rangle = 2\langle S(\mathbf{a}), S(\mathbf{b})\rangle$$

Orthogonal Linear Maps on $\mathbb{R}^n$

Orthogonal maps are associated with a unique orthogonal matrix

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be an isometry. Since T is linear, it is determined by its basis and there is a unique $Q \in \mathbb{R}^{n\times n}$ associated with T.

T is orthogonal so $\langle T_Q(\mathbf{x}), T_Q(\mathbf{y})\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$. But $\langle T_Q(\mathbf{x}), T_Q(\mathbf{y})\rangle = \langle Q\mathbf{x}, Q\mathbf{y}\rangle = (Q\mathbf{x})^T(Q\mathbf{y}) \Rightarrow (Q\mathbf{x})^T(Q\mathbf{y}) = \mathbf{x}^T\mathbf{y}$, thus

$$\mathbf{x}^T(Q^TQ)\mathbf{x} = \mathbf{x}^T\mathbf{y}$$

Setting $Q\mathbf{e}_i = \mathbf{q}_i \Rightarrow \mathbf{q}_i^T\mathbf{q}_j = \mathbf{e}_i^T\mathbf{e}_j$ and so $Q^TQ = I$.

Working backwards, we see the orthogonal matrix defines an isometry.

This means $O(n, \mathbb{R})$ the orthogonal group is also the group of isometries of $\mathbb{R}^n$.

Projections

$$P_{\mathbf{a}}(\mathbf{x}) = \frac{\langle \mathbf{a}, \mathbf{x}\rangle}{\langle \mathbf{a}, \mathbf{a}\rangle}\mathbf{a}$$

is the *projection* onto $\mathbb{R}\mathbf{a}$, and is linear.

When $\mathbf{a} = \mathbf{u}$ is a unit vector then we see that

$$P_{\mathbf{u}}(\mathbf{x}) = \mathbf{u}^T x\mathbf{u}$$

Multiplying this out

$$\begin{aligned}P_{\mathbf{u}}\begin{pmatrix}x_1\\x_2\end{pmatrix} &= (u_1x_1 + u_2x_2)\begin{pmatrix}u_1\\u_2\end{pmatrix} \\ &= \begin{pmatrix}u_1^2 & u_1u_2\\u_1u_2 & u_2^2\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}\end{aligned}$$

This projection matrix of $P_{\mathbf{u}}$ is symmetric. $\mathrm{im}(P_{\mathbf{u}}) = \mathbb{R}\mathbf{u}$, while $\ker(P_{\mathbf{u}})$ is the line orthogonal to $\mathbb{R}\mathbf{u}$.

A projection has the property that $P\mathbf{u} \circ P_{\mathbf{u}} = P_{\mathbf{u}}$.

$$P_{\mathbf{u}} \circ P_{\mathbf{u}}(\mathbf{v}) = \langle \mathbf{u}, \langle \mathbf{u}, \mathbf{v}\rangle\mathbf{u}\rangle\mathbf{u} = \langle \mathbf{u}, \mathbf{v}\rangle\mathbf{u} = P_{\mathbf{u}}(\mathbf{v})$$

1. P_W is linear
2. $P_W(\mathbf{w}) = \mathbf{w}$ if $\mathbf{w} \in W$
3. $P_W(W^\perp) = \{\mathbf{0}\}$
4. $P_W + P_{W^\perp} = I$

Reflections

Let $\mathbf{u} \in \ell^\perp$ and decompose $\mathbf{v} = P_{\mathbf{u}}(\mathbf{v}) + \mathbf{c}$ with $c \in \ell$.

$$\begin{aligned} H(\mathbf{v}) &= -P_{\mathbf{u}}(\mathbf{v}) + \mathbf{c} \\ &= -P_{\mathbf{u}}(\mathbf{v}) + (\mathbf{v} - P_{\mathbf{u}}(\mathbf{v})) \\ &= \mathbf{v} - 2P_{\mathbf{u}}(\mathbf{v}) \end{aligned}$$

Since $P_{\mathbf{u}}$ is a linear map, so H is a linear map.

Reflection Matrix

$$H(\mathbf{v}) = \mathbf{v} - 2P\mathbf{v}$$

$$Q = (I - 2P) = \begin{pmatrix} u_2^2 - u_1^2 & -2u_1u_2 \\ -2u_1u_2 & u_1^2 - u_2^2 \end{pmatrix}$$

using the identity $u_1^2 + u_2^2 = 1$

We can see this has the form $Q = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ where $a^2 + b^2 = 1$ and is a symmetric orthogonal matrix.

$$Q^2 = (I - 2P)^2 = I - 4P + 4P^2 = I$$

since $P^2 = P$. Since Q is symmetric, $Q = Q^T$ is therefore orthogonal.

$O(2, \mathbb{R})$-dichotomy

$$\det : O(2, \mathbb{R}) \to \mathbb{R}^\times$$

$$Q^TQ = I \Rightarrow \det(Q) = \pm 1$$

$$Q^TQ = \begin{pmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{pmatrix} = I$$

```
sage: var("a b c d")
(a, b, c, d)
sage: Q = matrix([[a, b], [c, d]])
sage: (Q.transpose() * Q).determinant().expand()
b^2*c^2 - 2*a*b*c*d + a^2*d^2
sage:
sage: c = -b*d/a
sage: d2 = a^2 + b^2 - c^2
sage: d2 = d2.expand()
sage: d2
a^2 + b^2 - b^2*d^2/a^2
sage: e = d^2 - d2 == 0
sage: e
-a^2 - b^2 + b^2*d^2/a^2 + d^2 == 0
sage: e -= b^2*d^2/a^2 + d^2
sage: e
-a^2 - b^2 == -b^2*d^2/a^2 - d^2
sage: e /= (-b^2/a^2 - 1)
sage: e.simplify_full()
a^2 == d^2
```

Either $a = d$ or $a = -d$.

$\ker(\det) = \mathrm{Rot}(2)$

Let $a = d \Rightarrow b = -c$ and $Q_1 = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$.

Then $\det(Q_1) = a^2 + b^2 = 1$. But this is the form of the rotation matrices $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

$O(2,\mathbb{R}) - \mathrm{Rot}(2)$ are Reflections

Let $a = -d \Rightarrow b = c$ and $Q_2 = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$. Rewriting it

$$Q_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} a & c \\ c & -a \end{pmatrix}$$

So Q_2 is a reflection and a rotation (coset of Q_1).

$\mathbf{v} = (-c \qquad a+1)^T, \mathbf{w} = (a+1 \qquad c)$ are orthogonal, and $Q\mathbf{v} = -\mathbf{v}$, $Q\mathbf{w} = \mathbf{w}$. Normalize them to get an orthonormal basis.

Exercise 7.3.9: Projection on Subspace Without Orthonormal Basis

W is a subspace with $\dim W = n$ of $\mathbb{R}^m$ with a basis $A = (\mathbf{a}_1 \cdots \mathbf{a}_n)$.

Let $\mathbf{x} \in \mathbb{R}^n$, when is $A\mathbf{x}$ the projection of $\mathbf{x}$ on W?

$$\mathbf{x} = \langle \mathbf{a}_1, A\mathbf{x} \rangle + \cdots + \langle \mathbf{a}_n, A\mathbf{x} \rangle \Rightarrow A^T A\mathbf{x} = \mathbf{x}$$

Prove $A^T A$ is invertible

Because we're working in $\mathbb{R}$, all dot products are non-negative

$$\langle Ax, Ax \rangle \geq 0$$

with the only zero being when $A\mathbf{x} = \mathbf{0}$. But $\dim \mathrm{col}(A) = n \Rightarrow \dim \ker(A) = 0 \Rightarrow \mathbf{x} = \mathbf{0}$.

Thus $\ker(A^T A) = \{\mathbf{0}\}$.

Let $P = A(A^T A)^{-1} A^T$. Prove $P\mathbf{x} \in W$ if $\mathbf{x} \in \mathbb{R}^m$, and $P\mathbf{x} = \mathbf{x}$ if $\mathbf{x} \in W$

$$T_A : \mathbb{R}^n \to \mathbb{R}^m$$

$$T_{A^T} : \mathbb{R}^m \to \mathbb{R}^n$$

$$T_{(A^T A)} : \mathbb{R}^n \to \mathbb{R}^n$$

$$T_{(A^T A)^{-1} A^T} : \mathbb{R}^m \to \mathbb{R}^n$$

Let $\mathbf{y} = T_{(A^T A)^{-1} A^T}(\mathbf{x}) \in \mathbb{R}^n$, then since $\mathrm{col}(A) = W$, and $\forall \mathbf{v} \in \mathbb{R}^n, A\mathbf{v} \in W$, so $A\mathbf{y} \in W$.

We showed in the previous part that $\ker(A^T A) = \{\mathbf{0}\} \Rightarrow T_{(A^T A)}$ is an isomorphism with inverse $T_{(A^T A)^{-1}} : \mathbb{R}^n \to \mathbb{R}^n$ so

$$T_{(A^T A)^{-1}} T_{(A^T A)} = T_{(A^T A)^{-1}} T_{A^T} T_A = e$$

which means that

$$(A^T A)^{-1} A^T A = I_n$$

but $A = (\mathbf{a}_1 \cdots \mathbf{a}_n)$ so we can left multiply by A

$$(A^T A)^{-1} A^T A = (\mathbf{e}_1 \cdots \mathbf{e}_n)$$

$$(A^T A)^{-1} A^T \mathbf{a}_i = \mathbf{e}_i \Rightarrow A(A^T A)^{-1} A^T \mathbf{a}_i = \mathbf{a}_i$$

which when viewing $\mathbf{x} = r_1 \mathbf{a}_1 + \cdots + r_n \mathbf{a}_n \in W$, we prove the result.

$\mathbf{v} - P\mathbf{v}$ is orthogonal to $W, \forall \mathbf{v} \in \mathbb{R}^m$

We can extend the basis $A = (\mathbf{a}_1 \cdots \mathbf{a}_n)$ of W to a basis of $\mathbb{R}^m$.

Let $\mathbf{v} = P\mathbf{v} + \mathbf{c}$. We see $P\mathbf{v} \in W \Rightarrow \mathbf{c} \notin W$. Since $\mathbb{R}^m = W \oplus W^\perp$, so $\mathbf{c} \in W^\perp$ and $\langle \mathbf{a}_i, \mathbf{c} \rangle = \mathbf{0}$ but $\mathbf{c} = \mathbf{v} - P\mathbf{v}$ and $A^T \mathbf{c} = \mathbf{0}$ is equivalent to $\begin{pmatrix} \langle \mathbf{a}_1, \mathbf{c} \rangle \\ \vdots \\ \langle \mathbf{a}_n, \mathbf{c} \rangle \end{pmatrix} = \mathbf{0}$.

Exercise 7.3.10

$$Q = \begin{pmatrix} a & b \\ b & -a \end{pmatrix} : a^2 + b^2 = 1 \sim Q = \begin{pmatrix} 1-2u_1^2 & -2u_1u_2 \\ -2u_1u_2 & 1-2u_2^2 \end{pmatrix} : u_1^2 + u_2^2 = 1$$

$$X = \bar{K}[u_1, u_2]/\langle u_1^2 + u_2^2 - 1\rangle$$

$$Y = \bar{K}[a, b]/\langle a^2 + b^2 - 1\rangle$$

$$\phi : X \to Y$$

$$\phi(u_1, u_2) = (-u_1^2 + u_2^2, -2u_1u_2)$$

We prove ϕ is an isomorphism.

Surjective: $b = 0 \Rightarrow$ either u_1 or u_2 is 0. But also $a = 0 \Rightarrow u_2$ or u_1 is 0. Thus $\ker \phi = \{\mathbf{0}\}$.

Injective:

$$\begin{aligned} a &= -u_1^2 + u_2^2 \\ &= (u_1^2 + u_2^2 - 1) + 1 - 2u_1^2 \\ &= 1 - 2u_1^2 \\ \Rightarrow u_1^2 &= \frac{1-a}{2} \\ u_2 &= -\frac{b}{2u_1} \end{aligned}$$

Show Q is a reflection through $\mathbb{R}\begin{pmatrix} -u_2 \\ u_1 \end{pmatrix}$.

$$\mathbf{v} = \begin{pmatrix} -u_2 \\ u_1 \end{pmatrix}$$

We want $\mathbf{w}$ such that $\langle \mathbf{v}, \mathbf{w}\rangle = 0 \Rightarrow \mathbf{w} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$.

Calculating, we see $Q\mathbf{v} = \mathbf{v}$ and $Q\mathbf{w} = -\mathbf{w}$, so Q is the reflection around $\mathbf{v}$.

$\mathcal{M}_B^B(T \circ S) = \mathcal{M}_B^B(T)\mathcal{M}_B^B(S)$

$$T(\mathbf{v}_j) = \sum_{i=1}^{n} c_{ij}\mathbf{v}_i \quad \mathcal{M}_B^B(T) = (c_{ij})$$

$$S(\mathbf{v}_j) = \sum_{i=1}^{n} d_{ij}\mathbf{v}_i \quad \mathcal{M}_B^B(T) = (d_{ij})$$

$$\begin{aligned} T(S(\mathbf{v}_j)) &= d_{1j}T(\mathbf{v}_1) + \cdots + d_{nj}T(\mathbf{v}_n) \\ &= d_{1j}(c_{11}\mathbf{v}_1 + \cdots + c_{n1}\mathbf{v}_n) + \cdots + d_{nj}(c_{1n}\mathbf{v}_1 + \cdots + c_{nn}\mathbf{v}_n) \\ &= (\mathbf{v}_1 \cdots \mathbf{v}_n) \begin{pmatrix} d_{1j}c_{11} + \cdots + d_{nj}c_{1n} \\ \vdots \\ d_{1j}c_{n1} + \cdots + d_{nj}c_{nn} \end{pmatrix} \end{aligned}$$

$$\begin{aligned} (\mathcal{M}_B^B(T \circ S))_{ij} &= d_{1j}c_{i1} + \cdots + d_{nj}c_{in} \\ &= c_{i1}d_{1j} + \cdots + c_{in}d_{nj} \end{aligned}$$

$$\Longrightarrow \mathcal{M}_B^B(T \circ S) = \mathcal{M}_B^B(T)\mathcal{M}_B^B(S)$$

Exercise 7.4.4: $P_{\mathbf{a}}$ is Semisimple

$$r = \langle \mathbf{a}, \mathbf{a} \rangle \quad P_{\mathbf{a}} : \mathbb{R}^2 \to \mathbb{R}^2$$

$$\langle \mathbf{a}, \mathbf{x} \rangle = a_1 x_1 + a_2 x_2$$

$$P = \frac{1}{r} \begin{pmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{pmatrix}$$

$$T(\mathbf{u}_1) = \lambda_1 \mathbf{u}_1$$

$$T(\mathbf{u}_2) = \lambda_2 \mathbf{u}_2$$

$$u_{11} \begin{pmatrix} a_1^2 \\ a_1 a_2 \end{pmatrix} + u_{21} \begin{pmatrix} a_1 a_2 \\ a_2^2 \end{pmatrix} = \lambda_1 \begin{pmatrix} u_{11} \\ u_{21} \end{pmatrix}$$

Let $(u_{11}, u_{21}) = (a_1, a_2)$

$$\begin{aligned} a_1^3 + a_1 a_2^2 &= \lambda_1 a_1 \\ a_1^2 a_2 + a_2^3 &= \lambda_1 a_2 \\ \implies \lambda_1 &= a_1^2 + a_2^2 \end{aligned}$$

Now $\mathbf{u}_2 = (-a_2, a_1)$ and $\langle \mathbf{u}_1, \mathbf{u}_2 \rangle = 0$. Let $(u_{12}, u_{22}) = (-a_2, a_1)$

$$u_{12} \begin{pmatrix} a_1^2 \\ a_1 a_2 \end{pmatrix} + u_{22} \begin{pmatrix} a_1 a_2 \\ a_2^2 \end{pmatrix} = \lambda_2 \begin{pmatrix} u_{12} \\ u_{22} \end{pmatrix}$$

$$\begin{aligned} -a_1^2 a_2 + a_1^2 a_2 &= -\lambda_2 a_1 = 0 \\ -a_1 a_2^2 + a_1 a_2^2 &= -\lambda_1 a_2 = 0 \\ \implies \lambda_2 &= 0 \end{aligned}$$

so P is diagonal with basis $\{\mathbf{u}_1, \mathbf{u}_2\}$

$$\mathcal{M}_B^B(P_{\mathbf{a}}) = \begin{pmatrix} a_1^2 + a_2^2 & 0 \\ 0 & 0 \end{pmatrix}$$

Dual Space

$$V^* = L(V, \mathbb{F})$$

$$\mathbf{v}_i^*(\mathbf{v}_j) = \delta_{ij}$$

If $\mathbf{v}_1, ..., \mathbf{v}_n$ is a basis of V, then $\mathbf{v}_1^*, ..., \mathbf{v}_n^*$ is a basis of V^*

$$f(\mathbf{x}) = a_1 \mathbf{v}_1(\mathbf{x}) + \cdots + a_n \mathbf{v}_n(\mathbf{x}) = \mathbf{0}(\mathbf{x})$$

Evaluating this on the basis of V, we see

$$f(\mathbf{v}_i) = a_j \text{ and } \mathbf{0}(\mathbf{v}_i) = 0 \implies a_j = 0$$

$\Phi(\mathbf{v}) = \varphi_{\mathbf{v}}$ is an Isomorphism

Let V be a finite-dimensional inner product space over $\mathbb{R}$.

$$\varphi_{\mathbf{v}} : V \to \mathbb{R}$$

$$\varphi_{\mathbf{v}}(\mathbf{x}) = \langle \mathbf{v}, \mathbf{x} \rangle$$

$$\Phi : V \to V^*$$

$$\Phi(\mathbf{v}) = \varphi_{\mathbf{v}}$$

Since $\dim V = \dim V^*$, we only have to show Φ is injective.

$\Phi(\mathbf{v}) = \mathbf{0} \implies \langle \mathbf{v}, \mathbf{x} \rangle = 0$ for all $\mathbf{x}$ including $\mathbf{x} = \mathbf{v}$. So $\ker \Phi = \{\mathbf{0}\}$.

Characterization of the Determinant

For any $\sigma \in S(n), \mathcal{D}(C_{\sigma(1)}, ..., C_{\sigma(n)}) = \operatorname{sgn}(\sigma)\mathcal{D}(C_1, ..., C_n)$

Decompose σ into a product of transpositions and apply $\mathcal{D}(AE) = -\mathcal{D}(A)$.

Equivalence with $\det(C)$

Let $\mathbf{c}_j = \sum_i b_{ij}\mathbf{a}_i$ where $A = (\mathbf{a}_1 \cdots \mathbf{a}_n), C = (\mathbf{a}_1 \cdots \mathbf{a}_n)$, then applying multilinearity, we see

$$\mathcal{D}(C) = \left(\sum_{\sigma \in S(n)} \operatorname{sgn}(\sigma) b_{\sigma(1),1} \cdots b_{\sigma(n),n} \right) \mathcal{D}(A)$$

just by expanding out each column of C. Then set $A = I$ and get the desired result.

This also gives us the product rule for determinants since $C = AB$.

Exercise 7.5.1: Isomorphism $\Delta : V \to V^{**}$

$$\mathbf{v} \in V, \quad f_{\mathbf{v}} = \Delta(\mathbf{v}) \in V^{**}, \quad f_{\mathbf{v}}(\varphi) = \varphi(\mathbf{v})$$

Now let $\mathbf{w} \in \ker \Delta$, then $g = \Delta(\mathbf{w})$ and $g(\varphi) = \varphi(\mathbf{w}) = 0$ for all $\varphi \in V^*$.

Since $\mathbf{w}$ can be written in terms of a basis, let $x_1, ..., x_n \in V^*$ be the respective component functions so $x_1(\mathbf{w}) = \cdots = x_n(\mathbf{w}) = 0$ and we see $\mathbf{w} = \mathbf{0}$ so we prove injectivity.

$$\dim L(V, \mathbb{F}) = n = \dim V$$

so $\dim V^{**} = \dim V$, and surjectivity follows from the above on Δ being injective.

Lastly since $V^* = L(V, \mathbb{F})$ consists of linear maps with additive property

$$\begin{aligned} \Delta(\mathbf{v} + \mathbf{w})(\varphi) &= \varphi(\mathbf{v} + \mathbf{w}) \\ &= \varphi(\mathbf{v}) + \phi(\mathbf{w}) \\ &= \Delta(\mathbf{v})(\varphi) + \Delta(\mathbf{w})(\varphi) \end{aligned}$$

proving Δ is a homomorphism.

Exercise 7.5.2: Adjoint Map $T^* : W^* \to V^*$

$$T^*(\omega) = \omega \circ T$$

T^* is Well-Defined and Linear

$W^* = L(W, \mathbb{F})$, and $T : V \to W$, so it follows their composition is well defined and linear.

Matrix of T^* wrt Dual Bases

$$T(\mathbf{v}_1) = a_{11}\mathbf{w}_1 + \cdots + a_{m1}\mathbf{w}_m$$
$$T^*(\mathbf{w}_1^*)(\mathbf{v}_1) = \mathbf{w}_1^*(T(\mathbf{v}_1)) = a_{11}$$
$$\cdots$$
$$T^*(\mathbf{w}_m^*)(\mathbf{v}_1) = a_{m1}$$
$$\cdots$$
$$T^*(\mathbf{w}_1^*)(\mathbf{v}_n) = a_{1n}$$
$$\cdots$$
$$T^*(\mathbf{w}_m^*)(\mathbf{v}_n) = a_{mn}$$

$$T^*(\mathbf{w}_1^* \cdots \mathbf{w}_m^*) = (\mathbf{v}_1^* \cdots \mathbf{v}_n^*)B$$

$$T^*(\mathbf{w}_1^*)(\mathbf{v}_1) = a_{11} \quad T^*(\mathbf{w}_1^*)(\mathbf{v}_n) = a_{1n}$$
$$\implies T^*(\mathbf{w}_1^*) = a_{11}\mathbf{v}_1^* + \cdots + a_{1n}\mathbf{v}_n*$$
$$B = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ & \vdots & \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} = A^T$$

If T is injective, then T^* is surjective

$$f \in V^* \quad T^*(\omega) = \omega \circ T$$

we prove there exists $\omega \in W^*$ such that $f = \omega T \in V^*$.

Since f is linear, it suffices to prove $\mathbf{v}_i^* \in \operatorname{im} T^*(W^*)$ for a single basis element of V^*.

Let $V^\perp$ be the space orthogonal to $\mathbf{v}_i$. Then because T is injective, $T\mathbf{v}_i \notin V^\perp$. Therefore we can define ω such that $\omega(T\mathbf{v}_i) = 1$ and $\omega(T\mathbf{v}_j) = 0$ for all $j \neq i$.

But then since $f(\mathbf{v}_i) = \omega \circ T(\mathbf{v}_i) = 1$, and $f(\mathbf{v}_j) = 0$ for $j \neq i$, we have $f = \mathbf{v}_i^*$.

If T is surjective, then T^* is injective

Let $T^*(z_1) = T^*(z_2)$, then $z_1 T = z_2 T \in V^*$.

Since T is surjective, both $z_1, z_2 \in W^* = L(W, \mathbb{F})$ are defined over all of W, so $z_1(\mathbf{w}) = z_2(\mathbf{w}) \quad \forall \mathbf{w} \in W \implies z_1 = z_2$.

$\dim \operatorname{im}(T) = \dim \operatorname{im}(T^*)$

$$\dim(T) = \dim(A) = \dim(A^T) = \dim(T^*)$$

Natural Surjective Linear Map $S : W^* \to V^*$ for $V \subseteq W$

$$S(\mathbf{w}^*)(\mathbf{v}) = \mathbf{w}^*(\mathbf{v})$$

Write $\mathbf{w} = \mathbf{v} + \mathbf{c}$, then $S(\mathbf{w}^*) = S(\mathbf{v}^*) + S(\mathbf{c}^*)$, which for all $\mathbf{x} \in V, S(\mathbf{w}*)(\mathbf{x}) = \mathbf{v} * (\mathbf{x})$